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SECTOR AT A GLANCE

Livestock

Agriculture and Allied Sectors in India

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The **'Sector-at-a-Glance'** provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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Overview

Livestock rearing is an integral part of agriculture and plays a significant role in fulfilling the nutritional and livelihood needs of the nation. The sector contributes almost 6% to national income and provides livelihood opportunities to more than two-thirds of the rural population.¹ Meat and poultry products constituted 6.4% of the total principal agricultural commodities exported in 2021-22.

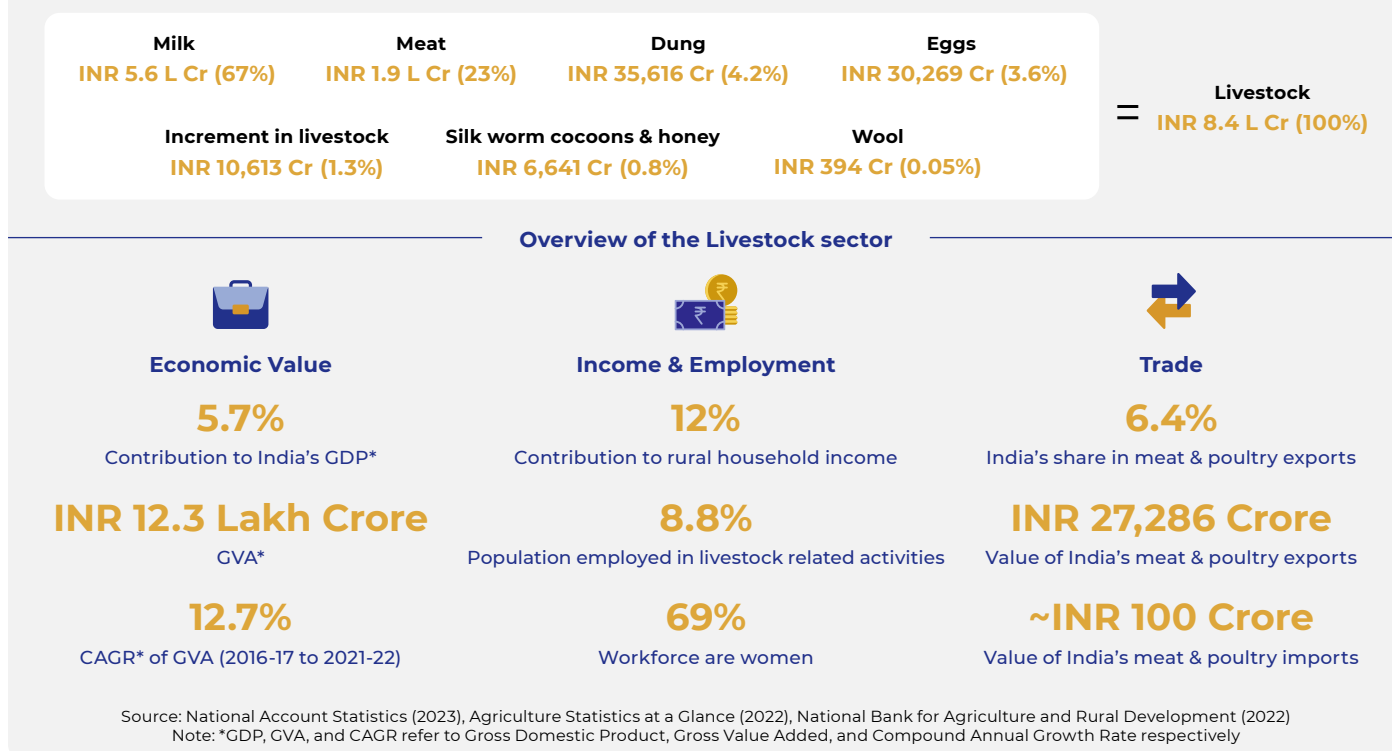
Enhancing livestock productivity and health, advanced livestock management and extension services, and upgradation of the meat value chain system are the key policy focus areas in the sector. These are reflected in the numerous initiatives undertaken by the Government to improve the productivity of the livestock sector and make it more sustainable - National Livestock Mission, Livestock Health and Disease Control, National Action Plan for Egg and Poultry, Integrated Cold Chain, and Value Addition Infrastructure among others.

Economic Value

India's livestock sector valued at INR 12.3 lakh crore contributed to 30.2% of the overall agricultural Gross Value Added (GVA) in 2021-22,² making it the second largest contributor to the agriculture and allied sector after field crops. The total value of output from the livestock sector is INR 8.4 lakh crore in 2021-22 with a compounded annual growth rate (CAGR) of 5.5% since 2016-17.³

Value addition in the livestock sector is largely driven by milk (67%) followed by meat (23%) and eggs (4%) sub-sectors. The insights here focus on eggs, meat, and wool sub-sectors.⁴

Figure 1: Overview of Sectoral Composition and Economic Contribution of the Livestock sector in India



1 [National Institute of Agricultural Extension Management. \(2021\). Livestock Extension Services: Time to Think Beyond Treatment and Breed Improvement.](#)

2 [Ministry of Agriculture and Farmers Welfare. \(2023\). Agriculture Statistics at a Glance 2022.](#)

3 [Ministry of Statistics and Programme Implementation. \(2023\). National Accounts Statistics 2023 — Statement 8.2.](#)

4 A separate Sector-at-a-glance document covers the dairy sub-sector

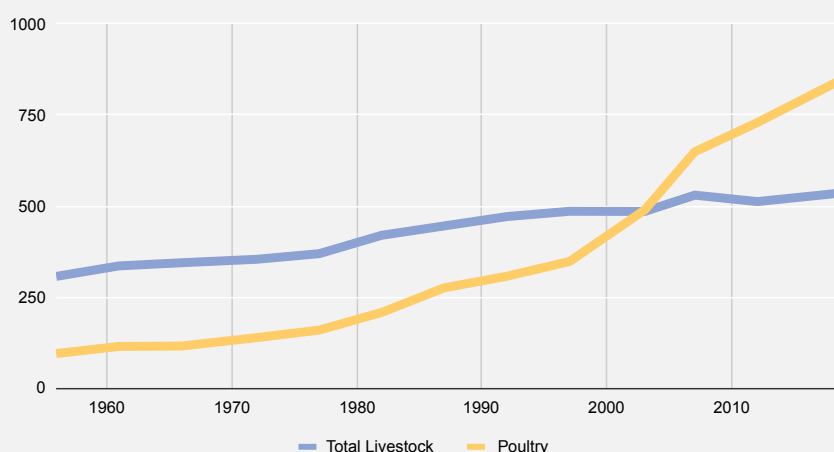
Population and Employment

India's livestock sector is one of the largest in the world¹ and is a major source of employment to small and marginal farmers, especially women. As per the latest Livestock Census, the total livestock population in India was 536.76 million in 2019.⁵

Livestock and Poultry Population Trend in India (1956-2019)

- Cattle (36%), goats (28%) and buffaloes (20%) constitute the majority share of the livestock population in India.
- The total livestock and poultry population has grown at an average annual growth rate of 0.9% and 3.5% respectively (Figure 2).

Figure 2: Livestock and Poultry Population Trend in India (1956-2019)

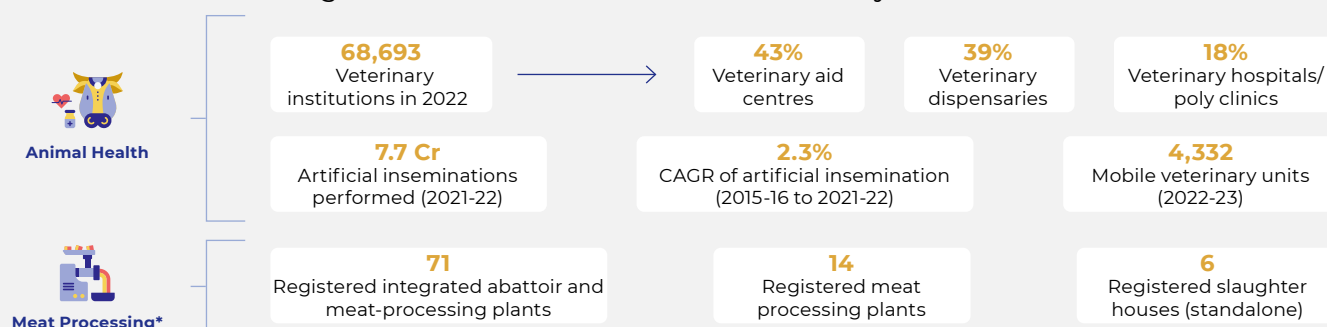


Source: Basic Animal Husbandry Statistics 2022

It is estimated that about 70 million rural households own livestock and about 20.5 million people depend upon it for their livelihood. This sector provides employment to about 8.8% of India's population,⁶ with women constituting about 69% of the workforce.⁷

Animal Health and Meat Processing Infrastructure^{8, 9, 10}

Figure 3: Infrastructure of Animal Husbandry Sector in India



Note: *Data reported as on 2 August, 2023

Source: Basic Animal Husbandry Statistics (2022), Department of Animal Husbandry and Dairying Annual Report 2022-23, Agricultural and Processed Food Products Export Development Authority (2023)

⁵ Department of Animal Husbandry & Dairying. (2019). *20th Livestock Census - 2019*.

⁶ National Bank for Agriculture and Rural Development. (2022). *Sustainable Meat Value Chain and Enhanced Farmers' Income*.

⁷ Department of Animal Husbandry, Dairying & Fisheries. (2013). *National Livestock Policy*.

⁸ Department of Animal Husbandry and Dairying. (2023). *Basic Animal Husbandry Statistics 2022*.

⁹ Department of Animal Husbandry and Dairying. (2023). *Annual Report 2022-23*.

¹⁰ Agricultural and Processed Food Products Export Development Authority. (2023) *List of Indian Integrated Abattoirs & Meat Processing Plants Approved*. Retrieved 25 September 2023.

Livestock exports, valued at INR 27,286 crores, contributed to 6.4% of overall agricultural exports in 2022-23. During the same period, the import of livestock products such as sheep/goat meat and poultry products as a percentage of total agricultural imports was negligible.¹¹

Sub-sectoral Trends

This section focuses on trends in production and international trade¹² of the following livestock sub-sectors:

- Eggs
- Meat
- Wool

a. Eggs

India is the third largest egg-producing country in the world. In 2021-22, its total egg production was recorded at 130 billion with an average annual growth rate of 8% since 2016-17. Almost 85% of the country's egg production is undertaken by organised commercial farms while the remaining 15% is from a traditional system of rearing poultry birds in small numbers in the backyards of rural households.

Production

130 Bn Nos.

Total egg production in 2021-22

8%

CAGR (2017-22)

3rd

India's rank in global egg production

85%

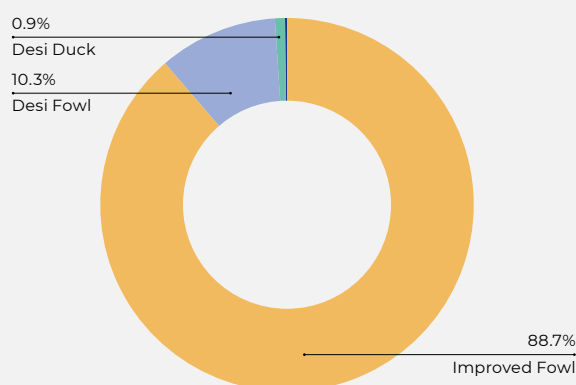
Egg production from commercial poultry

15%

Egg production from backyard poultry

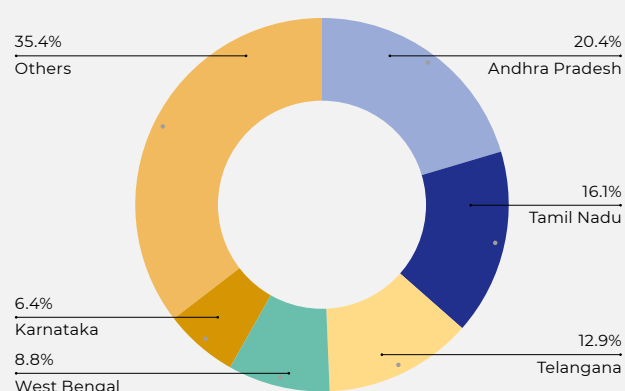
Majority of the egg production (90%) is by improved fowl species while the remaining 10% is from desi fowl, desi duck and improved duck species (Figure 4). The top five egg-producing states together contribute to 65% of the total egg production in the country: Andhra Pradesh is the leading egg producer (20%), followed by Tamil Nadu (16%), Telangana (13%), West Bengal (9%), and Karnataka (6%) (Figure 5).

Figure 4: Species-wise Egg Contribution (2021-22)



Source: Basic Animal Husbandry Statistics (2022)

Figure 5: Top 5 Egg-producing States in India (2021-22)



Source: Basic Animal Husbandry Statistics (2022)

¹¹ Trade data on livestock excludes dairy products

¹² Basic Animal Husbandry Statistics 2022 and Agricultural and Processed Food Products Export Development Authority (APE-DA) are the primary sources of data used in this document for production and international trade statistics respectively for the livestock sub-sectors.

The per capita availability of eggs was 95 eggs per annum in 2021-22 with an annual average growth rate of 7% since 2016-17. In 2021-22, the per capita availability of eggs among the top 15 states ranged from 501 eggs per annum (Andhra Pradesh) to 45 eggs per annum (Uttarakhand).

Trade

Poultry products include eggs, live poultry meat, and edible poultry meat. India's export of poultry products was valued at INR 1,082 crores (0.3% of total agricultural exports) in 2022-23 which has grown at an annual average rate of 14% since 2017-18. The country's import of poultry products constituted only 0.02% of total agricultural imports during the same period.

b. Meat

India is the eighth largest meat producer in the world. In 2021-22, the total meat production in the country was recorded at 9.3 million metric tonnes (MT) with an average annual growth rate of 4.7% since 2016-17. India is the largest producer of buffalo meat and the second largest producer of goat meat globally.⁹

Production

9.3 Mn MT

Total meat
production in 2021-22

4.7%

CAGR (2017-22)

8th

India's rank in global
in meat production
CAGR (2017-22)

1st

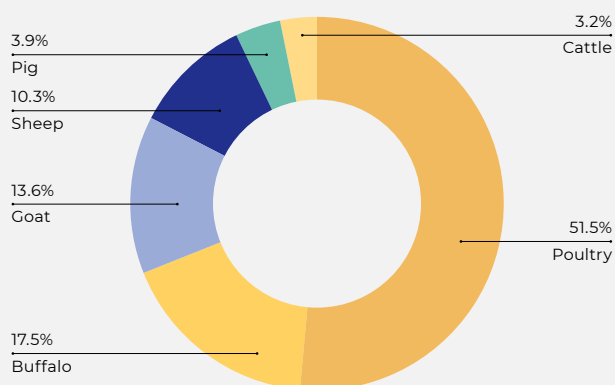
India's rank in
global buffalo meat
production

2nd

India's rank in global
goat meat production

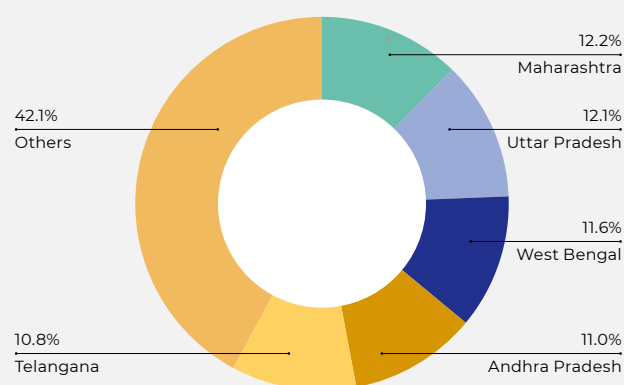
More than half (51%) of the meat produced in the country is poultry followed by buffalo, goat, sheep, pig, and cattle (Figure 6). Maharashtra is the top meat-producing state in India (12.2%) (Figure 7). The top five meat-producing states together contribute to 58% of the total meat production in the country.

Figure 6: Species-wise Meat Contribution (2021-22)



Source: Basic Animal Husbandry Statistics (2022)

Figure 7: Top 5 Meat-producing States in India (2021-22)



Source: Basic Animal Husbandry Statistics (2022)

The per capita availability of meat was 6.8 kg/annum in 2021-22 and has grown at an annual average rate of 3.6% since 2016-17. In 2021-22, the per capita availability of meat among the top 15 states ranged from 26.6 kg/annum (Telangana) to 8.9 kg/annum (Tamil Nadu).

Trade

Buffalo meat is the most exported meat product. In 2022-23, buffalo meat exports were valued at INR 25,640 crores (6% of total agricultural exports) and had almost nil growth rate since 2017-18. The import of poultry products, sheep/goat meat, and other meat accounts for less than 0.1% (in value terms) of total agricultural imports in 2022-23. **India is the fourth largest buffalo meat exporter in the world with Malaysia, Vietnam, Egypt, Indonesia and Iraq being the key export destinations in 2022-23.**¹³

c. Wool

India has the third largest sheep population (74.26 million sheep) and is the ninth largest wool-producing country in the world.¹⁴ In 2021-22, the total wool production in the country was recorded at 33 million kg which has decreased at an annual rate of 5.3% since 2016-17.

Production

33 Mn Kg

Total wool production
(2021-22)

- 5.3%

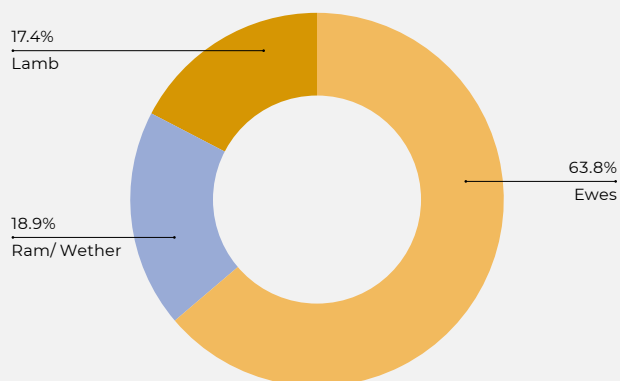
CAGR (2017-22)

9th

India's rank in global
wool production

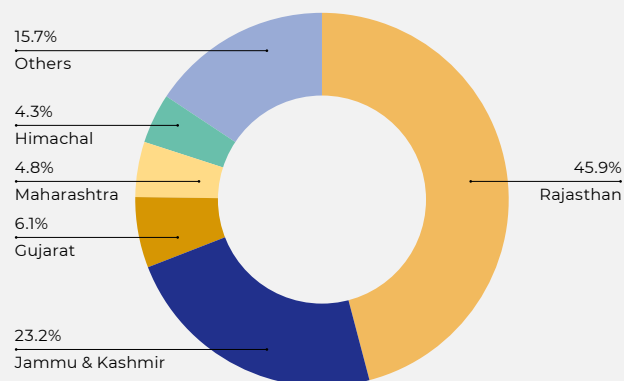
The majority (64%) of the wool production is contributed by ewes followed by lamb and ram species. Rajasthan is the top wool-producing state in India (Figure 8). The top five wool-producing states together contribute to 84% of the total wool production in the country (Figure 9). Rajasthan is the top producer of wool, contributing 46% to national production.

Figure 8: Species-wise Wool Contribution
(2021-22)



Source: Basic Animal Husbandry Statistics (2022)

Figure 9: Top 5 Wool-producing States in India
(2021-22)



Source: Basic Animal Husbandry Statistics (2022)

Trade

Raw wool and woollen items such as yarn, fabrics, readymade garments and carpets are the major items traded in the international market. India is a major importer of raw wool. From 2016-17 to 2020-21, the average quantity and value of raw wool imported were 79 million kg and INR 1705 crores respectively. Australia, New Zealand, China, and Turkey are the major countries from which India imports raw wool.¹⁴

Carpets, valued at INR 10,846 crores, are the major woollen item exported from India with a market share of 87% in 2020-21. The United States of America is the primary export destination for Indian handmade carpets followed by Germany, Australia and the United Kingdom.¹⁴

¹³ Agricultural and Processed Food Products Export Development Authority. (n.d.). *Buffalo Meat*.

¹⁴ Ministry of Textiles. (2022). *Wool Sector at a Glance*.

Opportunities & Challenges

Opportunities



Meat production potential:

The Salvage of Male Buffalo Calves (SMBC) is promoted through the Entrepreneurship Development and Employment Generation (EDEG) scheme – a component of the Government of India's National Livestock Mission (NLM). It is estimated that nearly 10–12 million male buffalo calves are removed from the production system due to their non-remunerative raising costs. Salvaging these calves from early death and growing them to a 250 kg live weight would result in the production of one million MT of boneless high-quality buffalo meat. Export-oriented slaughterhouses should establish backward integration with farmers for raising these animals by providing health, feeding, and extension management services at farmers' doorsteps and buying them back at remunerative prices.¹⁵



Demand-supply gap:

The Indian Council of Medical Research's expert committee recommends 60 g of protein per day with a net protein utilisation (NPU) of 65g. To achieve this NPU, a minimum requirement of 20g per capita per day is required. The Food and Agriculture Organization (FAO) projects that the national poultry meat consumption will double by 2030, necessitating increased meat production to meet national nutritional security.¹⁵



Livestock as a sustainable source of livelihood for rural population:

Livestock contributes to ~15-20% of farmer's annual household incomes in India.¹⁶ As the distribution of land holdings in rural India is highly skewed towards medium and large farmers, rearing of livestock, particularly small ruminants, pigs, and poultry, is considered the most sustainable income option for poor households.



Adoption of modern technology:

Due to the growing consumer demand for clean, safe and hygienic meat, and meat and poultry products, adopting modern technology such as farm automation, abattoirs, meat-processing plants and cold chain infrastructure in India will prove valuable.¹⁷



Changing face of Indian meat marketing:

Indian meat marketing is evolving with the establishment of modern retail outlets in Tier-I and Tier-II cities. This shift aims to make fresh, marinated, and ready-to-eat meat products more attractive to consumers. Online meat marketing start-ups are also growing in these cities, offering differentiated products and transforming the inefficient supply chain into an efficient platform. The companies prominently focusing on meat supply include Licious, Farmage, Meatigo, Starchick, Fresho, Fresh To Home, TenderCuts, Neatmeat, and Zappfresh.¹⁵

¹⁵ Muthukumar et al. (2021). *An overview of Indian livestock and meat sector*.

¹⁶ Directorate of Economics and Statistics. (2019). *Debt and Investment, Land & Livestock Holdings of Households and Situation Assessment of Agricultural Households in Delhi*.

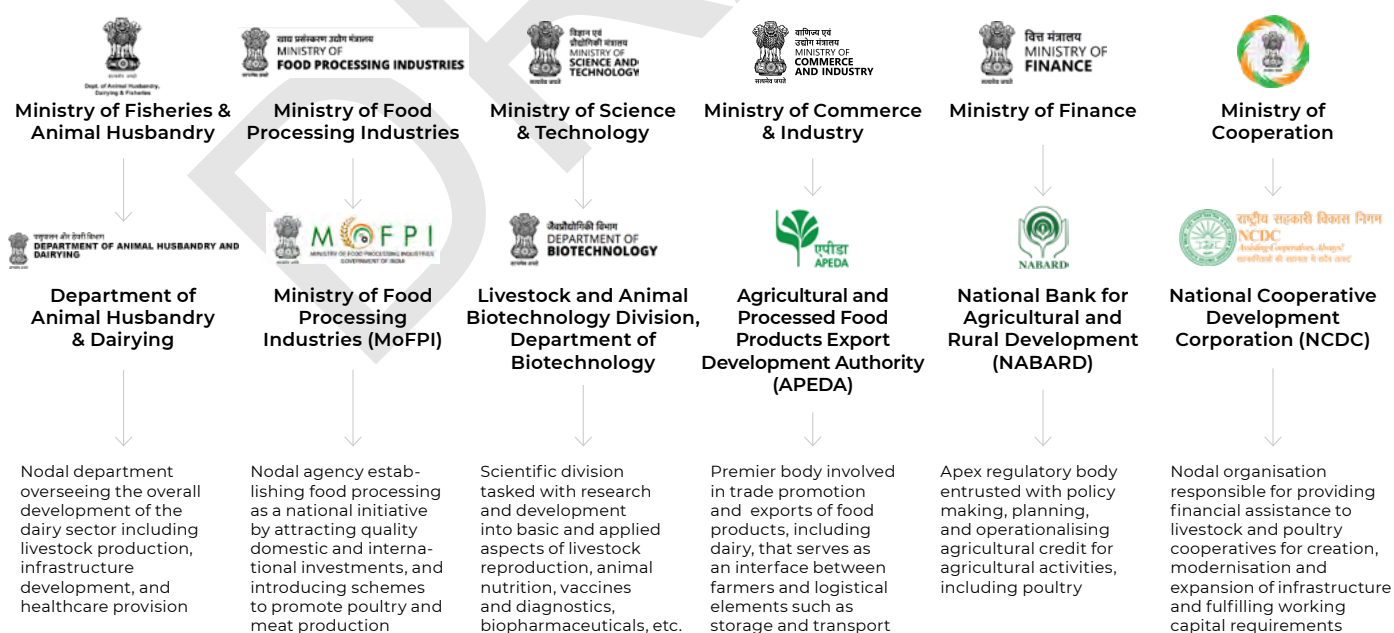
¹⁷ Ministry of Food Processing Industries. (2016). *Opportunities in Meat & Poultry Sector In India*.

Challenges

 Sectoral Challenges	• Poor uptake of technology resulting in low meat production • Under-utilisation of byproducts and improper waste disposal • Lack of prioritisation of extension services like providing technical services to animals and educating livestock farmers • Need for greater investments (the sector received only ~12% of the total public expenditure of agriculture and allied sectors)¹ • Non-availability of superior pedigree males for breeding
 Input-level Challenges	• Non-availability of quality feed and fodder • Poor transportation and water/feeding conditions affecting overall meat quality • Lack of access to animal-feed-testing laboratories in India • Inadequate coverage of vaccination resulting in an economic loss of INR 50,000 crores to farmers due to animal diseases¹ • Lack of infrastructure and human resources for support services
 Production-level Challenges	• Frequent outbreaks of diseases like Foot and Mouth Disease, Black Quarter infection, Influenza etc. affecting livestock health and lower productivity • Inadequate infrastructure facilities for clean meat production and processing (e.g. slaughterhouses lack basic facilities for hygienic slaughter) • Inadequate modernisation of district/regional laboratories and veterinary hospitals/dispensaries across India
 Post-production-level Challenges	• Inadequate marketing infrastructure (e.g. retail shops lack basic facilities for the hygienic handling of meat which influences consumers' purchasing behaviour) • Inadequate cold chain and transport systems • Lack of rigorous compliance by registered meat processing plants and abattoirs with global food safety and quality requirements

Government Interventions

Ministries and Departments





SECTORAL INTERVENTIONS

National Action Plan for Egg & Poultry

Established in 2022 to promote farmer incomes and fulfill the Indian population's protein requirements

National Livestock Mission

Launched for promoting breed improvement and entrepreneurship development through three sub-missions

Livestock Census and Integrated Sample Survey

Initiated to supply animal husbandry statistics and estimations of livestock products

Poultry Venture Capital Fund

Introduced to ensure the sustainable development of livestock

National Digital Livestock Mission

Implemented to increase productivity, control diseases, and ensure quality of livestock.

Integrated Wool Development Programme

Aims to enhance wool marketing and processing, as well as human resource capacities, in the wool sector



INPUT-LEVEL INTERVENTIONS

Animal Husbandry Startup Grand Challenge

Launched to locate innovative and commercially-viable solutions to address problems faced by the animal husbandry sector

Animal Husbandry Infrastructure Development Fund

Implemented to develop bovine breeding in a scientific manner to improve cow productivity and milk production

National Animal Disease Control Programme

Launched to control Foot and Mouth disease and Brucellosis through appropriate vaccination



PRODUCTION-LEVEL INTERVENTIONS

Livestock Health and Disease Control Scheme

Initiated to promote the vaccination of livestock to prevent and control the spread of diseases

National Livestock Mission

Policy framework for improving the productivity of the livestock sector in a sustainable manner

National One Health Mission

Intends to coordinate overall health emergency preparedness and unified disease control against important human and animal diseases, with alerts and rapid response actions built on integrated surveillance



POST PRODUCTION-LEVEL INTERVENTIONS

Integrated Cold Chain and Value Addition Infrastructure

Provides integrated and continuous cold chain and preservation infrastructure facilities, from the farm to the consumer

Pradhan Mantri Kisan Sampada Yojana

Established to make provisions for value addition through mega food parks and ensure cold chain infrastructure

Production Linked Incentive Scheme for Food Processing Industry

Supports Indian food product brands in international markets

 Increased focus on enhancing livestock productivity and health	Breed improvement measures • Propagate the use of high genetic merit bulls for breeding purposes via the Rashtriya Gokul Mission • Train manpower to conduct artificial insemination services at farmers' doorsteps in rural areas via the Multi Purpose Artificial Insemination Technicians in Rural India (MAITRI) initiative • Utilise whole genome sequencing and genomic chips for DNA-based selection of cattle breeds	Eradication of infectious diseases • Animal Pandemic Preparedness Initiative (APPI): Design an overall integrated disease surveillance and reporting system built on the National Digital Livestock Mission for the enhanced surveillance of zoonotic and other animal diseases and pandemic preparedness at the community-level • Scale the production of veterinary vaccines and drugs
 Advanced livestock management and sustainability	Artificial Intelligence (AI) and Automation systems for livestock farms • Precision Livestock Farming: Use Artificial Intelligence (AI) for effective health monitoring, data collection of individual animals and improvement in feed quality • Use AI in poultry farming to automate repetitive tasks such as collecting, counting and packaging eggs, which helps increase production, efficiency, accuracy, safety, and reduce labour and time requirements • Reduce the high installation and repair costs involved in farm automation to enable accessibility for non-commercial farmers	Sustainable livestock practices and upskilling of livestock farmers • Increase the population of indigenous cattle breeds which are more resilient to climate change and diseases compared to commercially bred cattle • Provide a more nutrient-rich diet to livestock during thermal stress to help minimise production losses • Livestock extension services: Provide proper and constant training of farmers regarding feeding, breeding, health, management, upgradation, and utilisation of new technologies via cooperative societies, Self Help Groups (SHGs) and Farmer Producer Organisations (FPOs).
 Upgradation of the meat value chain	Meat processing, cold chain infrastructure and market linkage • Increase the number of meat-processing plants and cold storage facilities • Strengthen the linkages between production and markets through institutions such as co-operatives, producers' associations, and contract farming	Food safety and quality assurance • Ensure stricter compliance of commercial feeds intended for meat and milk producing animals with the Bureau of Indian Standards (BIS) specifications • Prepare producers for quality-driven competition in the domestic as well as global market

¹⁸ Department of Animal Husbandry & Dairying. (2022). *Revised Operational guidelines for Implementation of Rashtriya Gokul Mission for 2021-2026*.

¹⁹ Pashudhan Praharee. (2022). *Artificial Intelligence (AI): Future of Livestock farming in India*. Retrieved 25 September 2023.

Recommended Readings

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